Uses of potassium permanganate:

Some important uses of potassium permanganate are listed below.

1. It is used as a strong oxidizing agent.
2. It is used for the treatment of various skin infections and fungal infections of the foot.
3. It used in water treatment industries to remove iron and hydrogen sulphide from well water.
4. It is used as a Bayer's reagent for detecting unsaturation in an organic compound.
5. It is used in quantitative analysis for the estimation of ferroussalts, oxalates, hydrogen peroxide and iodides.

HCl cannot be used for making the medium acidic since it reacts with $KMnO_4$ as follows.

$$2MnO_4^- + 10\,Cl^- + 16H^+ \longrightarrow 2Mn^{2+} + 5Cl_2 + 8H_2O$$

HNO_3 also cannot be used since it is good oxidising agent and reacts with reducing agents in the reaction.

However,H_2SO_4 is found to be most suitable since it does not react with potassium permanganate.

$$\text{Equivalent weight of } KMnO_4 \text{ in acid medium} = \frac{\text{Molecular weight of } KMnO_4}{\text{no of mols of electrons transferred}} = \frac{158}{5} = 31.6$$

$$\text{Equivalent weight of } KMnO_4 \text{ in basic medium} = \frac{\text{Molecular weight of } KMnO_4}{\text{no of mols of electrons transferred}} = \frac{158}{1} = 158$$

$$\text{Equivalent weight of } KMnO_4 \text{ in neutral medium} = \frac{\text{Molecular weight of } KMnO_4}{\text{no of mols of electrons transferred}} = \frac{158}{3} = 52.67$$

f-block elements – inner transition elements

In the inner transition elements there are two series of elements.

1) Lanthanoids (previously called lanthanoides)
2) Actinoids (previously called actinides)

Lanthanoid series consists of fourteen elements from Cerium ($_{58}$Ce) to Lutetium ($_{71}$Lu) following Lanthanum ($_{57}$La).These elements are characterised by the preferential filling of 4f orbitals, Similarly actinoids consists of 14 elements from Thorium ($_{90}$Th) to Lawrencium ($_{103}$Lr) following Actinium ($_{89}$Ac).These elements are characterised by the preferential filling of 5f orbital.

The position of Lanthanoids in the periodic table

The actual position of Lanthanoids in the periodic table is at group number 3 and period

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